Table 2 (scenario sim1, sim1_pilot, $B = 48$ per cell). Estimated exposure effects of cumulative total exposure with model misspecification (bold cells with correct model specification).

	Estimated β (true: Cox 0.4055, ERR 0.423)								Pairs
	True x		Naive		RRC corrected				dropped
$\rho(c, C)$	Cox	ERR	Cox	ERR	Cox.A	Cox.M	ERR.A	ERR.M	(%)
A. Mean $\hat{\beta}$ over converged fits									
Cox+Additive	Cox true $\beta = 0.4055$, additive independent error								
0.3	0.38	0.56	0.22	0.39	**0.35**	0.37	0.61	1.62	
0.6	0.42	0.72	0.27	0.51	**0.41**	0.46	0.77	1.93	
0.9	0.51	1.01	0.34	0.68	**0.51**	0.52	1.02	1.14	
Cox+Multiplicative	Cox true $\beta = 0.4055$, multiplicative independent error								
0.3	0.37	0.61	0.24	0.37	0.37	**0.37**	0.57	0.57	
0.6	0.36	0.55	0.24	0.38	0.35	**0.35**	0.55	0.55	
0.9	0.36	0.57	0.24	0.37	0.35	**0.36**	0.55	0.56	
ERR+Additive	ERR true $\beta = 0.423$, additive independent error								
0.3	0.32	0.44	0.21	0.31	0.31	0.35	**0.46**	1.88	
0.6	0.44	0.71	0.30	0.47	0.45	0.51	**0.71**	1.15	
0.9	0.38	0.65	0.26	0.44	0.38	0.40	**0.65**	0.73	
ERR+Multiplicative	ERR true $\beta = 0.423$, multiplicative independent error								
0.3	0.32	0.53	0.26	0.42	0.40	0.39	0.66	**0.63**	
0.6	0.34	0.56	0.25	0.40	0.38	0.37	0.62	**0.60**	
0.9	0.31	0.52	0.20	0.34	0.31	0.31	0.52	**0.52**	
B. Median $\hat{\beta}$ over all fits; [number of fits not converged or stopped at the cap $\beta = 50$]									
Cox+Additive	Cox true $\beta = 0.4055$, additive independent error								
0.3	0.32	0.37	0.26	0.32	**0.43**	0.44	0.51	0.55 [3]	28.4
0.6	0.43	0.54	0.24	0.30	**0.39**	0.46	0.43	0.61	12.4
0.9	0.56	0.78	0.36	0.51	**0.55**	0.57	0.78	0.79	4.3
Cox+Multiplicative	Cox true $\beta = 0.4055$, multiplicative independent error								
0.3	0.38	0.51	0.24	0.31	0.36	**0.39**	0.50	0.49	0.0
0.6	0.37	0.52	0.24	0.33	0.38	**0.36**	0.47	0.47	0.0
0.9	0.33	0.40	0.21	0.26	0.31	**0.31**	0.40	0.39	0.0
ERR+Additive	ERR true $\beta = 0.423$, additive independent error								
0.3	0.37	0.43	0.25	0.30	0.31	0.43	**0.42**	0.51 [1]	28.5
0.6	0.40	0.53	0.34	0.42	0.51	0.58	**0.68**	0.72	12.4
0.9	0.35	0.44	0.26	0.31	0.39	0.39	**0.46**	0.48	4.2
ERR+Multiplicative	ERR true $\beta = 0.423$, multiplicative independent error								
0.3	0.37	0.51	0.27	0.34	0.43	0.44	0.56	**0.52**	0.0
0.6	0.33	0.46	0.24	0.32	0.37	0.36	0.49	**0.50**	0.0
0.9	0.32	0.36 [1]	0.21	0.26 [1]	0.33	0.32	0.40 [1]	**0.39 [1]**	0.0

Cox model $\lambda(t) = \lambda_0(t)\exp(\beta x(t))$ with true $\beta_{\text{Cox}} = 0.4055$; ERR model $\lambda(t) = \lambda_0(t)(1 + \beta x(t))$ with true $\beta_{\text{ERR}} = 0.423$. Each fit is compared with the true β of the model it fits; the paper's values (for $\beta = 1$ and 1.109) are not shown. True x: the same risk model fitted to the true cumulative exposure, separating risk-model misspecification and small-sample bias from measurement error.

$n_1 = 2000$, $n_2 = 500$ (at most 8 pairs each), $T^* = 60$, cumulative incidence about 10%. Surrogate C has compound symmetry ($\rho = 0.5$) across years, $E(C) = 0.012$; true $c = \gamma_1 C + \epsilon$ with $\gamma_1 = 2/3$ (additive) or $\log c = \gamma_0 + \log C + \epsilon$ with $E(c) = 0.008$ (multiplicative), σ_ϵ set by $\rho(c, C)$; median x 0.21 (additive) and 0.18 (multiplicative).

Pairs dropped: share of validation pairs with $c \leq 0$ or $C \leq 0$, which the log (.M) calibration cannot use. – : no fit converged, or not recorded. A median of 50.00 means at least half of the fits stopped at the cap $\beta = 50$.